

Physical Activity Lead

Physical-activity prevalence

NATIONAL POPULATION HEALTH SURVEY 2024

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https://isomer-user-content.by.gov.sg/3/4cad4e62-8093-48de-b0d1-af8e012d4d95/NPHS%202024%20Survey%20Report_Final.pdf

Table 3.1: Age-specific prevalence (%) of sufficient total physical activity among Singapore residents aged 18 to 74 years by sex, 2024

Age (years)	Total	Males	Females
18-29	89.7	91.4	87.8
30-39	88.2	91.4	85.1
40-49	84.8	85.7	83.9
50-59	81.7	80.0	83.4
60-74	80.3	79.7	80.9
18-74	84.7	85.4	84.0

Among Singapore residents aged 60-74 (age band closest to WiSE dementia prevalence population) 80.3% met the recommended sufficient total physical activity level in 2025. So 19.7% were insufficiently active.

Risk estimate

PHYSICAL ACTIVITY AS A PROTECTIVE FACTOR FOR DEMENTIA AND ALZHEIMER'S DISEASE: SYSTEMATIC REVIEW, META-ANALYSIS AND QUALITY ASSESSMENT OF COHORT AND CASE-CONTROL STUDIES

Iso-Markku P, Kujala UM, Knittle K, Polet J, Vuoksima E, Waller K. Physical activity as a protective factor for dementia and Alzheimer's disease: systematic review, meta-analysis and quality assessment of cohort and case-control studies. Br J Sports Med. 2022 Jun;56(12):701-709. doi: 10.1136/bjsports-2021-104981. Epub 2022 Mar 17. PMID: 35301183; PMCID: PMC9163715.

<https://pubmed.ncbi.nlm.nih.gov/35301183/>

“PA was associated with a decreased risk of all-cause dementia (pooled relative risk 0.80, 95% CI 0.77 to 0.84, n=257 983), Alzheimer’s disease (0.86, 95% CI 0.80 to 0.93, n=128 261) and vascular dementia (0.79, 95% CI 0.66 to 0.95, n=33 870), even in longer follow-ups (≥ 20 years) for all-cause dementia and Alzheimer’s disease.”

Inactive individuals have a 25% higher dementia risk than active people ($1/0.8 = 1.25$)

Scenarios

- 15% reduction in physical inactivity (WHO’s target)

“The WHO *Global action plan on physical activity 2018–2030* provides policy guidance on how countries could support people of all ages and abilities to be more regularly active. In the World Health Assembly resolution (WHA71.6) supporting the action plan, all countries agreed to a global target of a 15% relative reduction (against a 2010 baseline) in insufficient physical activity by 2030.”

<https://iris.who.int/server/api/core/bitstreams/33339c9c-3a9f-46d4-9f12-ae9ff0dfdc6a/content>

- 5% reduction in physical inactivity (Impact National Step Challenge)

The NSC is a large-scale health initiative, aiming to increase physical activity levels among its participants through daily step and activity tracking. Participants are provided with free wearable devices and incentivised with rewards for meeting step-count goals. A simulation-based forecasting analysis from March 2026 projects that the NSC would lower obesity and T2D prevalence by 2050. The study does not estimate changes in physical activity or dementia, but we could use it to base our assumption of a (realistic) scenario on it. Taking the average enrolment rate of the NSC and assuming that 50% of the inactive participants become sufficiently active, gives a relative reduction in activity of about 5%.

The 50% conversion rate is our own assumption and it assumes that inactive adults enrol at the same rate as others which likely overstates the effect. We therefore use 5% as an optimistic low-impact scenario.

The study looks at the people aged 18-74.

Simulation scenario	Enrolment rate (%)	Retention rate (%)	Rationale
No NSC	0	0	Serves as a reference scenario to quantify the underlying burden of obesity and T2DM in the absence of any further population-wide physical activity intervention, enabling rigorous assessment of the incremental benefits of NSC.
Average NSC	9.6	28.3	Represents the expected impact of maintaining the NSC programme at mean historical enrolment and retention rates (NSC3-5), reflecting a realistic projection of current practice.
Peak NSC	11.7	32.4	Captures the programme's maximum observed rates (NSC1-5) by modelling outcomes at the highest historical enrolment and retention rates, illustrating the effects of optimal past implementation.
Double NSC	19.2	56.6	Assesses the potential impact of substantially scaling up the programme through intensified policy and community efforts, helping to estimate the benefits of ambitious but potentially achievable expansion.
Full NSC	100	100	Provides an estimate of the theoretical maximum public health benefit by modelling universal participation, thereby defining the upper limit of preventable disease burden achievable through physical activity interventions.

Table 1: Enrolment rate, retention rate and rationale for five simulation scenarios covering years 2022-2050.

<https://www.sciencedirect.com/science/article/pii/S2666606526000350>

- 50% reductions in physical inactivity

Using this scenario we could illustrate the preventable burden if physical inactivity were reduced substantially amongst older people. Moreover, we achieve a range of results from low to high.

Method

We estimate how much a reduction in physical inactivity could lower dementia cases using the. The inputs are the share of residents aged 60-74 who are insufficiently active (19.7%) and the relative risk of dementia among inactive compared to active people (1.25). For each scenario inactivity is reduced by a percentage.

Population Attributable Fraction (PAF).

$$PAF = \frac{P(RR - 1)}{P(RR - 1) + 1}$$

- P = share of the population who are inactive
- RR = relative risk — how many times more likely an inactive person is to develop dementia (roughly 1.4 in the literature)

Limitations

- NPHS does not consider age group 75+ while WiSE covers 80+

Prevalence of insufficient physical activity among adults aged 18+ years (age-standardized estimate) (%)

FILTERS

Last updated: 2026-09-04

Indicator Location	Prevalence of insufficient physical activity among adults aged 18+ years (age-standardized estimate) (%)		
	Both sexes	Male	Female
Singapore			
2022			
80+ years		44.2 [30.3-58.0]	51.3 [37.2-65.4]
70-79 years		35.0 [22.0-48.0]	35.6 [22.4-48.8]
60-69 years		28.7 [16.6-40.9]	25.0 [13.7-36.3]
18+ years	23.2 [17.2-29.9]	24.2 [15.8-34.0]	22.0 [13.7-31.8]
2021			
80+ years		43.9 [30.9-56.9]	51.2 [37.9-64.6]
70-79 years		35.0 [22.7-47.2]	35.8 [23.3-48.4]
60-69 years		28.6 [17.3-40.0]	25.1 [14.0-36.1]
18+ years	23.1 [17.6-29.3]	24.1 [16.2-33.0]	22.1 [14.3-31.1]
2020			
80+ years		44.0 [31.5-56.6]	51.4 [38.6-64.2]
70-79 years		34.8 [23.3-46.3]	35.9 [24.1-47.7]
60-69 years		28.5 [17.8-39.3]	25.1 [14.8-35.3]
18+ years	23.1 [17.9-28.9]	24.0 [16.7-32.2]	22.1 [14.9-30.6]
2019			
80+ years		43.9 [32.2-55.6]	51.5 [39.5-63.6]
70-79 years		34.7 [23.6-45.7]	36.0 [24.6-47.5]
60-69 years		28.5 [18.2-38.8]	25.1 [15.3-34.9]
18+ years	23.1 [18.3-28.3]	23.9 [17.2-31.4]	22.2 [15.5-29.9]
2018			
80+ years		43.8 [32.6-55.0]	51.6 [40.0-63.2]

Data WHO

[https://www.who.int/data/gho/data/indicators/indicator-details/GHO/prevalence-of-insufficient-physical-activity-among-adults-aged-18-years-\(age-standardized-estimate\)-\(-\)](https://www.who.int/data/gho/data/indicators/indicator-details/GHO/prevalence-of-insufficient-physical-activity-among-adults-aged-18-years-(age-standardized-estimate)-(-))

- RR assumed constant